Comparative Analysis of Knowledge Graphs and Vector Databases for Enhancing Factual Accuracy in Retrieval-Augmented Generation (RAG) Systems

Overview

This report provides a technical comparison between Knowledge Graphs (GraphRAG) and Vector Databases (Standard RAG) for improving factual accuracy in Retrieval-Augmented Generation (RAG) systems. Vector databases perform approximate nearest neighbor search over dense embeddings, excelling at capturing semantic similarity and unstructured contexts. Conversely, knowledge graphs organize information explicitly as entities and relations, enabling multi-hop reasoning, structural precision, and deterministic relational retrieval. Recent benchmarking highlights the tradeoffs between structural fidelity and unstructured coverage.

Comparison

Criterion	Approach A	Approach B
Multi-hop Reasoning and Relational Queries	Knowledge graphs inherently support multi-hop reasoning by traversing explicit edges between entities, making them highly effective for queries requiring connected contextual information across distant document sections.	Vector databases struggle with strict multi-hop relational traversal because similarity search retrieves chunks independently based on local semantic proximity rather than global structural links.
Handling Unstructured Text and Semantic Similarity	Knowledge graphs can lose nuanced narrative flow, contextual subtleties, and verbatim descriptive text during the entity and relation extraction process.	Vector databases excel at indexing raw unstructured text, capturing subtle semantic nuances, and retrieving relevant passages using dense neural embeddings.
Factual Hallucination and Precision	Graph-based RAG architectures reduce certain types of hallucinations by grounding generation in explicit, verifiable	Standard vector RAG systems are prone to retrieving semantically related but factually irrelevant or misleading passages, which can

	relational triples and structured schemas.	induce hallucinations in downstream LLMs.
Construction and Maintenance Complexity	Knowledge graphs require complex data pipelines involving Named Entity Recognition (NER), relation extraction, schema design, and entity resolution, which can be brittle and expensive.	Vector databases are straightforward to set up, requiring only text chunking, embedding generation via pre-trained encoder models, and approximate nearest neighbor indexing.

Strengths & Weaknesses

Approach A

+ Enables complex multi-hop reasoning and structural global search over interconnected entities.
+ Provides explicit provenance and relational paths to mitigate factual hallucinations.
+ Integrates naturally with structured domain ontologies and deterministic rules.
- High extraction complexity and susceptibility to information loss during entity/relation parsing.
- Computationally intensive graph construction and expensive query-time traversal for large-scale datasets.
- Rigid schema definitions can hinder adaptability when domain data evolves rapidly.

Approach B

+ Seamlessly ingests unstructured text without requiring manual schema definitions or complex NLP extraction pipelines.
+ Highly scalable vector indices (e.g., HNSW, IVF) allow fast approximate nearest neighbor retrieval across millions of documents.
+ Captures broad semantic intent and paraphrased queries effectively.
- Lacks explicit structural awareness, leading to failures in multi-hop or relationship-heavy queries.
- Prone to retrieving noisy or top-k semantically similar chunks that lack precise factual answers.
- Vulnerable to semantic drift where embeddings map close vectors to unrelated facts.

Recommendation

> For applications requiring precise relational tracking, multi-hop logical traversal, and strict factual constraints (e.g., legal compliance or biomedical discovery), Knowledge Graphs (GraphRAG) are recommended. For general-purpose document search and semantic matching over vast unstructured corpora, Vector Databases are preferable. In many high-performance production

> systems, a hybrid approach combining vector search for localized semantic recall and knowledge graphs for structured relational reasoning yields optimal factual accuracy.

Sources

Do We Still Need GraphRAG? Benchmarking RAG and GraphRAG for Agentic Search Systems
Provides comparative benchmarks evaluating the performance of RAG and GraphRAG systems in agentic search workflows.
https://arxiv.org/abs/2604.09666v1

When to use Graphs in RAG: A Comprehensive Analysis for Graph Retrieval-Augmented Generation
Analyzes the specific conditions and use cases where incorporating knowledge graphs into RAG pipelines significantly improves accuracy.
https://arxiv.org/abs/2506.05690v3

GraphRAG under Fire
Discusses limitations, failure modes, and trade-offs associated with graph-based retrieval augmented generation architectures.
https://arxiv.org/abs/2501.14050v4